# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

**First/Second Semester of B. Tech. A.T.K.T. Examination (EC/ME/CL/CE/IT/EE)**

**December 2011**

**EE 101 Fundamentals of Electrical & Electronics Engineering (Old Course)**

**Date: 08.12.2011, Thursday** **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

***Instructions:***

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

## SECTION - I

**Q - 1 (a)** What is coefficient of coupling? Derive an expression for the same. **[05]**

**(b)** State Faraday's laws of electromagnetic induction. **[02]**

**Q - 2 (a)** A ring has mean diameter of 15 cm, a cross-section of 1.7 $cm^2$ and has a radial gap of 0.5 mm cut in it. It is uniformly wound with 1500 turns of insulated wire and a current of 1 A produces a flux of 0.1 mWb across the air gap. Calculate the relative permeability of iron on the assumption that there is no magnetic leakage. **[05]**

**(b)** Derive an expression of energy stored in capacitor of 'C' farad when voltage 'V' is applied across it. **[05]**

**(c)** A 20 μF capacitor initially charged to a p.d. of 500 V is discharged through an unknown resistance. After one minute, the p.d. at the terminals of the capacitor is 200 V. what is the magnitude of the resistance? **[04]**

**OR**

**Q - 2 (a)** Explain the following terms :- **[04]**
(1) Electric field intensity (2) Permittivity (3) Electric flux density (4) Electrical Potential

**(b)** A coil which has an initial temperature of 20 °C is connected to a 180 V supply and the initial current is found to be 4 A. Sometimes later the current is found to have decreased to 3.4 A, the supply voltage being unchanged at 180 V. Determine the temperature rise of the coil. Assume $\propto_0 = 0.0043\ {}^0C^{-1}$. **[04]**

**(c)** Calculate the force on a charge of $+10^{-2}$ C placed at a point (0, 0) due to point charges $Q_1$ and $Q_2$. The charge $Q_1$= $+10^{-6}$C is at (-2, 2) meters and $Q_2$= -2 x $10^{-6}$C is at (2, 2) meters. **[06]**

**Q - 3 (a)** Derive an expression of current and power factor for R-L-C series circuit connected to single phase a.c. supply. Also draw vector diagram for the circuit with (1) $X_L>X_C$, (2) $X_L<X_C$ and (3) $X_L=X_C$. **[07]**

**(b)** A series circuit has resistance of 10 ohms, inductance ( $^{200}/_{\pi}$) mH and capacitance ( $^{1000}/_{\pi}$) μF. Calculate: (i) the current flowing in the circuit of supply voltage 250 V, 50 Hz. (ii) power factor of the circuit (iii) power drawn from the supply. **[03]**

(c) Define following terms: [04]
[a] Amplitude
[b] Power Factor
[c] R.M.S value
[d] Crest factor

**OR**

Q - 3 (a) Derive an expression for the instantaneous value of alternating sinusoidal emf in terms of its maximum value, angular frequency and time. [07]

(b) Drive the relationship between line voltage and phase voltage, line current and phase current in case of 3 phase star connected system. [03]

(c) A capacitor of 10μF is connected across 230 V, 50 Hz supply. Find the capacitive reactance and the current drawn from the supply. What alternation in current drawn will occur if frequency is (1) doubled and (2) reduced to half? The supply voltage is kept constant in each case. [04]

## SECTION – II

Q - 4 (a) Draw and explain the *V-I* characteristics of PN junction. Also explain the operation of full-wave Bridge rectifier. [07]

Q - 5 (a) Explain how transistor uses as an amplifier? Derive the relationship between α and β. [07]

(b) Explain different filters circuits in detail. [07]

**OR**

Q - 5 (a) Write in brief for following terms: [06]
a. Breakdown voltage
b. Knee voltage
c. Forward current
d. Leakage current
e. Peak Inverse Voltage
f. Ripple factor

(b) A crystal diode having internal resistance $r_f$ =20 Ω is used for half-wave rectification. If the applied voltage v=50sinωt and load resistance $R_L$= 800 Ω find: [04]

a. $I_m$, $I_{dc}$, $I_{rms}$
b. d.c. output voltage
c. efficiency of rectification

(c) Describe the transistor action in detail. [04]

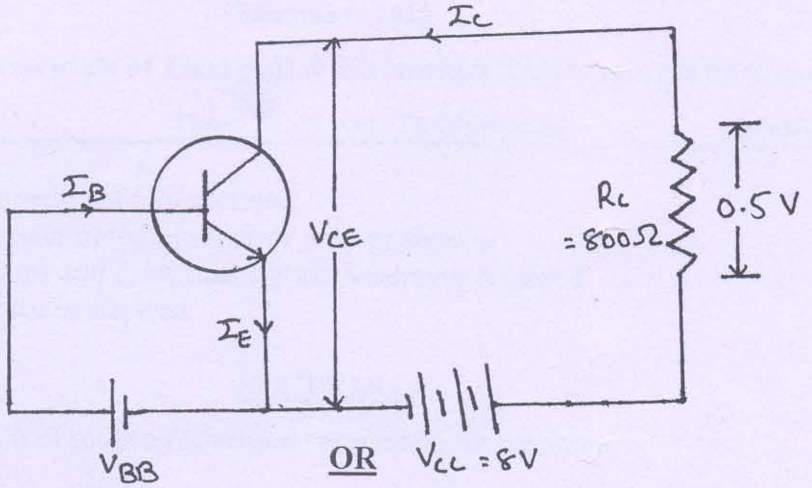
Q - 6 (a) (I) Explain any three terms [07]
(a) rise time, (b) fall time, (c) periodic signal, (4) duty cycle

(II) (a) Perform $(5174)_8$ - $(7425)_8$ using 8's complement. (b) Perform $(11001)_2$ - $(01110)_2$

(b) (i) Convert (1110011011)2 number into octal number. [03]
(ii) Convert (ADE7)16 number into decimal number.

(c) A transistor is connected to a CE configuration where collector supply voltage is 8V and voltage drop across resistor Rc is 0.5V. Take α=0.96 and find [04]

(i) collector-emitter voltage (ii) base current




**OR**

**Q - 6 (a)** Why voltage stabilizer circuit is required? How Zener diode is work as Voltage Regulator. **[07]**

**(b)** (i) Give in brief about operating point. **[07]**
(ii) calculate emitter current in a transistor for which $\beta = 40$ and base current = 15μA
(iii) find the $\alpha$ if (a) $\beta$=90, (b) $\beta$=40

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